Demo 1 — Federated Medical Action Recognition

Privacy-Preserving Fall Detection via Skeleton-Based FL

Pipeline Overview

FL Training (Colab T4)

NTU RGB+D 60 dataset
↓ filter 10 classes
5 hospital clients
(non-IID, Dirichlet)
STGCN++ (443K params)
50 rounds FedAvg + BN

Patient data never leaves
the hospital (FL)

ONNX
→
export

Real-Time Inference (CPU)

Webcam / Video input
↓
YOLOX → RTMPose → STGCN
↓
10 medical actions
15-20 FPS, CPU only

Only skeleton keypoints —
no face/identity on camera

10 Focused Classes

#	Medical Actions (7)	#	Normal Context (3)
0	**Falling**	7	Standing up
1	Staggering	8	Sitting down
2	Touch head (headache)	9	Walking towards
3	Touch chest (heart pain)		
4	Touch back (backache)		
5	Touch neck (neckache)		
6	Nausea / vomiting		

Filtered from NTU-60's 56K samples → focused medical subset

FL Training Setup

Component	Detail
Data	NTU RGB+D 60 — HRNet 2D skeletons, 10-class subset
Non-IID split	Dirichlet α=0.5 across 5 clients
Warm-start	5 epochs centralized pretraining
FL rounds	50 rounds FedAvg + BN calibration
LR schedule	0.01 → 0.001 (R20) → 0.0001 (R35)
Model	STGCN++ — 6 stages, 443K params
ONNX export	~2 MB, verified against PyTorch

Privacy: raw skeleton data stays on each client — only model weights are aggregated

Deployment Stack

Model	Role	Size
`yolox_tiny.onnx`	Person detection	19 MB
`rtmpose_m.onnx`	17-keypoint pose	52 MB
`stgcnpp_medical_federated.onnx`	**FL medical recognition**	**~2 MB**

Input: `(1, 2, 100, 17, 3)` → **Output:** `(1, 10)` action logits

Runtime: `numpy` + `opencv` + `onnxruntime` — no PyTorch, no GPU needed

Training Outputs

The Colab notebook generates:

- **Non-IID data distribution heatmap** — 5 clients × 10 medical classes
- **4-panel FL analysis** — timeline, client heatmap, loss curves, summary
- **Confusion matrix** with per-class accuracy bars
- **Fall detection metrics** — recall, precision, F1, missed-fall analysis

Live Demo

Demo Commands

Our medical FL model (10 classes, fall detection):

```
python demo/demo_onnx.py --device cpu --threads 4 --fast \
    --short-side 640 \
    --recog-model stgcnpp_medical_federated.onnx \
    --label-map demo1_fed_skeleton/label_map_medical.txt
```

Pre-recorded video analysis (auto-splits into 3s windows):

```
python demo/demo_onnx.py --mode clip --clip test/video.mp4 \
    --device cpu --threads 4 \
    --recog-model stgcnpp_medical_federated.onnx \
    --label-map demo1_fed_skeleton/label_map_medical.txt
```

Pretrained baseline (120 classes, comparison):

```
python demo/demo_onnx.py --device cpu --threads 4 --fast --short-side 640
```